ImpactForge

The AI-Powered Student Philanthropy Engine

Hacktoberfest Hackathon NUST × NSTP 2026 | MIT License | Next.js · Claude AI · Stripe

United Nations Sustainable Development Goals

ImpactForge directly contributes to the following UN SDGs by lowering the barriers to charitable giving for university students and making philanthropic impact transparent and measurable:

SDG 1 – No Poverty: ImpactForge enables fundraising campaigns for flood relief, winter clothing, and community support in Pakistan's most underserved regions (Tharparkar, Dadu, Kech), channelling student donations directly toward poverty alleviation and disaster recovery for vulnerable populations.

SDG 4 – Quality Education: The platform hosts campaigns such as "Textbooks for Tribal Schools" that fund educational materials for children in remote areas, helping ensure inclusive and equitable quality education and promoting lifelong learning opportunities for all.

SDG 6 – Clean Water and Sanitation: The "Thar Water Wells" campaign raises funds for solar-powered water pumps and RO filtration systems in Tharparkar's desert villages, directly advancing universal access to safe and affordable drinking water for communities where women walk 6 km daily to reach brackish wells.

SDG 9 – Industry, Innovation and Infrastructure: ImpactForge uses AI (Anthropic Claude structured outputs) to auto-generate complete fundraising campaigns from a single sentence, innovating on traditional charity workflows with parallel AI generation, deterministic impact tracking, and a cinematic donor-experience pipeline.

SDG 17 – Partnerships for the Goals: The project is fully open-source (MIT License), built for the Hacktoberfest community, and designed for collaborative contribution. It bridges student societies, NGOs, and donors through a shared digital platform, strengthening partnerships for sustainable development.

1. Problem Statement

Campus charity drives at Pakistani universities run on WhatsApp forwards and Google Forms. Organizers — typically student society members — lose days writing copy, designing posters, and planning outreach with no professional tools. Donors rarely learn what their money actually did, eroding trust. **Low transparency leads to low repeat giving**, and promising initiatives stall after a single fundraiser. There is no streamlined, student-friendly platform that generates professional campaigns instantly, processes donations, and closes the loop with tangible impact stories.

2. Solution Overview

ImpactForge is an AI-powered philanthropy engine that turns a single sentence into a fully designed fundraising campaign — and then gives every donor a cinematic "Impact Wrapped" story of exactly where their money went. The platform has three pillars:

- **Campaign-in-a-Box:** An organizer types a cause (e.g., "winter coats for children in Kech"). Two parallel Claude AI calls generate themed landing-page copy, impact tiers with real PKR-to-outcome mappings, a transparent budget, FAQs, a WhatsApp share pitch, and a day-by-day launch strategy with KPIs — all in seconds.
- **Impact Wrapped:** After donating, every donor receives a swipeable 7-slide cinematic story featuring a Sankey diagram of fund allocation, an animated node graph (donor → campaign → beneficiaries), ripple effects, a delivery journey, and an exportable share card.
- **One-Tap Giving with Real Checkout:** A polished donation flow translates amounts into tangible outcomes, supports optional tips, fee coverage (so the cause nets 100%), monthly recurring giving, and sponsor matching. Stripe Checkout when configured; demo mode otherwise.

Additional features include a **Generation Theater** (watch the campaign assemble live), **generative cover art** (unique per campaign, no stock photos), a **WebGL globe** showing donation flows, an **organizer dashboard** (analytics, CSV export, billing), and **inline AI editing** with undo.

3. Technology Stack

Layer	Technologies
Framework	Next.js 16 (App Router, Cache Components), React 19, TypeScript (strict)
Styling	Tailwind CSS v4, CSS-variable design tokens, system SF Pro / Inter

Layer	Technologies
AI	Anthropic Claude API — structured outputs (messages.parse + zodOutputFormat), parallel generation, curated fallback
Payments	Stripe Checkout over REST (no SDK), HMAC-verified webhooks, demo fallback
Data Viz	Recharts (donut, Sankey), hand-built animated SVG node graph
Motion & UI	Motion (Framer Motion), Aceternity/Magic-UI components, canvas particles, Cobe WebGL globe
State	Zustand + localStorage persistence (skipHydration SSR-safe)
Validation	Zod v4 — two-layer schema: loose AI-facing → strict app schema
Quality	ESLint, Prettier, node:test unit tests, GitHub Actions CI
Dev Workflow	Kiro IDE: steering docs, EARS-format specs, agent hooks

4. Architecture

ImpactForge follows a **data-only AI pipeline**: the LLM returns structured JSON, never markup. The architecture flows as:

Student prompt → Prompt Studio → POST /api/generate (Zod input + rate limit) → Two parallel Claude calls (copy+theme, strategy) → Normalizer (clamp, sort, rebalance budget to 100%) → Strict CampaignSchema validation → Themed campaign page. If AI fails (no key, timeout, invalid output), the system falls back to curated keyword-matched campaigns or a template built from the prompt's own words. Donations flow through a PaymentProvider interface (mock or Stripe) into a seeded impact engine that generates deterministic Wrapped stories.

Key design decisions: constrained creativity (6 palettes × 4 hero styles × 4 moods), two-layer schemas to repair model drift, deterministic impact data keyed by donation ID, and swappable payment mocks behind a provider interface for future JazzCash/Easypaisa integration.

5. Setup Instructions

Prerequisites: Node.js 20+. An Anthropic API key is optional (the app runs in offline demo mode without one).

```
git clone https://github.com/your-username/impactforge.git
cd impactforge
npm install
cp .env.example .env.local  # add ANTHROPIC_API_KEY for live AI (optional)
npm run dev
```

Open http://localhost:3000, type a cause, and hit "Forge campaign". Set `STRIPE_SECRET_KEY` to enable real Stripe Checkout; set `DEMO_MODE=true` for curated campaigns on flaky Wi-Fi. Deploy to Vercel with one click — no database needed.

6. Team Contributions

Member	Role	Key Contributions
Team Lead	Full-stack & AI	AI generation pipeline (structured outputs, normalizer, fallback), Stripe integration, Kiro specs & steering, architecture design, CI/CD
Member 2	UI/UX & Motion	Campaign page design, Impact Wrapped slides, Generation Theater, generative cover art, globe animation, accessibility

7. Open-Source License & Collaboration

ImpactForge is distributed under the **MIT License**. The repository includes CONTRIBUTING.md (fork → branch → lint/typecheck/test/build → PR workflow), a Code of Conduct, SECURITY.md for vulnerability reporting, and THIRD_PARTY_NOTICES.md acknowledging all major dependencies. Issues are labelled *good first issue* and *hacktoberfest* to welcome community contributors.

8. Kiro IDE Usage

ImpactForge was built **spec-first** using the Kiro IDE. All Kiro artifacts are committed in the `.kiro/` directory:

- **Steering documents** (always-on rules): product scope & happy path, tech constraints (strict TS, no `any`, AI output is data not markup), project structure conventions, and a design-system spec (theme tokens, motion budget, mobile-first).

- **Specs** (EARS-format requirements): *Campaign-in-a-Box* (prompt input, parallel generation, theater, campaign page, mock donation), *Impact Wrapped* (7-slide cinematic story, deterministic data, share card), and *Monetization & Payments* (tip model, plans, Stripe integration, checkout math).
- **Agent hooks**: *seed-schema-guard* (re-validates all fallback campaigns when schema changes), *theme-token-lint* (flags hardcoded colors and motion ignoring reduced-motion), *ship-check* (one-click pre-demo lint + typecheck + test + build).

9. Demo

A live demo is deployed at **https://impactforge.vercel.app**. The repository includes a demo GIF (`public/demo.gif`) and nine annotated screenshots covering the landing page, Generation Theater, campaign page, checkout, receipt, Impact Wrapped, pricing, dashboard, and feature tiles. The demo happy path: Landing → type a cause → Forge campaign → Generation Theater → themed campaign page → Donate → confetti → Impact Wrapped (7 slides) → Dashboard.